Rohit Kumar

M.Sc. Physics Student (Astrophysics), Ariel University, Israel

rohitkumaragk@gmail.com | 21mph011@nith.ac.in | +91 8809343XXX
Website / GitHub | LinkedIn | Blog

Research Interests

Galaxy and star formation, Active Galactic Nuclei (AGN) feedback, black hole physics, and accretion in binary systems. I combine theoretical modeling with data analysis (X-ray and gamma-ray observations from NASA missions) and numerical methods in Python. I also have a long background in teaching and mentoring.

Education

- **M.Sc. Physics** Oct 2025 – Present
  Ariel University, Ariel, Israel
- **M.Sc. Physics** 2021 – 2023
  National Institute of Technology (NIT) Hamirpur, India CGPA: 7.52 / 10
- **B.Sc. Physics** 2016 – 2020
  Indira Gandhi National Open University (IGNOU), Delhi, India Percentage: 62%
- **Diploma in Elementary Education (D.El.Ed.)** 2017 – 2019
  G.J.P.H.S., Siwan, Bihar, India
- **Intermediate (Class XII)** 2016
  S.B.S. Inter College, Siwan, Bihar, India Percentage: 68.4%
- **Matriculation (Class X)** 2013
  G.J.P. High School, Siwan, Bihar, India Percentage: 74.4%

Research Experience

- **Accretion in Eccentric Binaries** 2025 – Present
  Ariel University, with Amit Kashi Comparing wind-fed and Roche-lobe-overflow (RLOF) accretion to determine when a transient accretion disk forms in eccentric binary systems. Results presented as a poster (see Presentations).
- **AGN and Blazar Study** July to Oct 2025
  BHU, India with Raj Prince Analysis of hard X-ray and multi-wavelength data from NASA missions (Swift BAT, XRT, UVOT, Fermi-LAT and NuSTAR) to study AGN and blazars.
- **Part-time Independent Researcher**
  Zooniverse Citizen-science research projects and data classification.

Conference Presentations

- **Poster (virtual):** *When Does a Transient Accretion Disk Form? Comparison of Wind-Fed and RLOF Accretion in Eccentric Binaries.* **Rohit Kumar** and Amit Kashi (Ariel University).
  2nd IAU–ORAS International Pro-Am Workshop, Oil Region Astronomy Learning Center, Pennsylvania, USA 7 October 2026

Technical Projects

- Construction of a 150 mm Newtonian reflecting telescope, NIT Hamirpur.
- Design and demonstration of a working Earth–Sun system model, NIT Hamirpur.

Schools, Workshops and Internships

- Webinar: *Color the Universe with Rubin Observatory* – AAS Feb 2025
- *Data Analysis in Cosmology using Python* – Sardar Patel University, Gujarat Oct 2024

- *Exploding Stars Investigation with Rubin Observatory* – Vera C. Rubin Observatory and AAS Jul 2024
- Summer School in Theoretical (Astro)Physics – St. Xavier's College Ahmedabad and IUCAA Jun 2024
- Student Internship – Indian Centre for Space Physics, Kolkata Nov 2022
- Workshop on Recent Progress in Astrophysics – Amity University, Greater Noida Nov 2022
- Summer Internship – IIIT Kota Jul 2021
- Summer School Internship – Indian Institute of Astrophysics (IIA)
- Workshop: *Theoretical and Data Analysis Techniques in Cosmology* – GHRCE Nagpur (IUCAA funded)
- Workshop: *Beginning Astronomy v2* – Central University of Himachal Pradesh and IUCAA

Teaching Experience

- **Tutor** 2019
  Coaching and Home Tuition, Delhi
- **Teacher** Apr 2015 – Mar 2019
  Adarsh Gyan Kendra, Barari, Siwan
- **Teacher** Apr 2015 – Mar 2019
  S.S.C. Coaching Centre, Sani Basantpur, Siwan

Skills

- **Programming and software:** Python, MATLAB, Origin, Linux, MS Office
- **Research:** Data analysis, statistical techniques, numerical analysis, scientific writing
- **Other:** Teaching, mentoring, communication, problem solving

Languages

Hindi (native), Bhojpuri (native), English (fluent; English course, Oxford School of English, Delhi)

References

Available on request.